

Proof for Lemmas in 1044

Efficient Multi-Sample Submatrix Parallel Framework for PageRank Approximation on Large Graphs

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1 Lemma 3.1 Sequential CUR-Trans

Lemma. 3.1 *Given $\theta \leq \text{rank}(A)$, let A_θ denote the optimal rank- θ approximation of A . If the column and row sampling ratio in CUR-Trans satisfies $\rho > \frac{2500\theta^2}{n}$, then the following bound holds with probability at least 0.9.*

$$\begin{aligned} \|A - CUR\|_F &\leq \sqrt{50}(1 - \rho n \alpha_c)^{1/4}(1 - \rho n \alpha_r)^{1/4} \|A\|_F \\ &\quad + \frac{50(1 - \rho n \alpha_r)^{1/2}(1 - \rho n \alpha_c)^{1/2}}{\sigma_{\min}(W)} \|A\|_F^2, \end{aligned} \quad (1)$$

where W is the intersection of matrices C and R , $\sigma_{\min}(W)$ is the smallest non-zero singular value W , $\alpha_r = \sum_i \|A_{i,:}\|_2^4 / \|A\|_F^4 \geq 1/n$, and $\alpha_c = \sum_j \|A_{:,j}\|_2^4 / \|A\|_F^4 \geq 1/n$. Consequently, the PageRank approximation error of CUR-Trans satisfies

$$\|\pi - \tilde{\pi}\|_2 \leq \frac{\alpha}{1 - \alpha} \|A - CUR\|_F. \quad (2)$$

Proof. For matrix $A \in \mathbb{R}^{n \times n}$, suppose that S_r is a matrix indicating that which rows are sampled in A , i.e., if row i is sampled, the $(S_r)_{i,i} = 1$, else $(S_r)_{i,i} = 0$, $R = S_r A$. The same for S_c is a matrix indicating that which columns are sampled in A , i.e., if column i is sampled, the $(S_c)_{i,i} = 1$, else $(S_c)_{i,i} = 0$, $C = A S_c$. Then, we have matrix $W = S_r A S_c$. Let the number of sampled columns and rows be c and r , respectively, and $c = r = \rho n$. Each row is sampled independently with probability p_i , and each column is sampled independently with probability q_j , where

$$p_i = \frac{\|A_{i,:}\|_2^2}{\|A\|_F^2}, \quad q_j = \frac{\|A_{:,j}\|_2^2}{\|A\|_F^2}.$$

Without loss of generality, we reorder the rows and columns such that sampled rows and columns appear first in A . Define $\mathcal{R}$ as the set of indices of the sampled rows, $\mathcal{C}$ as the set of indices of the sampled columns, $\mathcal{R}^c$ as the set of indices of the unsampled rows, and $\mathcal{C}^c$ as the set of indices of the unsampled columns.

$$A = \begin{bmatrix} W & A_{\mathcal{R}, \mathcal{C}^c} \\ A_{\mathcal{R}^c, \mathcal{C}} & A_{\mathcal{R}^c, \mathcal{C}^c} \end{bmatrix},$$

where $A_{\mathcal{R}, \mathcal{C}^c} = S_r A (I - S_c)$ is the intersection of sampled rows and unsampled columns, $A_{\mathcal{R}, \mathcal{C}^c} \in \mathbb{R}^{r \times (n-c)}$; $A_{\mathcal{R}^c, \mathcal{C}} = (I - S_r) A S_c$ is the intersection of unsampled rows and sampled columns, $A_{\mathcal{R}^c, \mathcal{C}} \in \mathbb{R}^{(n-r) \times c}$; and $A_{\mathcal{R}^c, \mathcal{C}^c} = (I - S_r) A (I - S_c)$ is the intersection of unsampled rows and unsampled columns, $A_{\mathcal{R}^c, \mathcal{C}^c} \in \mathbb{R}^{(n-r) \times (n-c)}$. Therefore,

$$R = S_r A = \begin{bmatrix} W & A_{\mathcal{R}, \mathcal{C}^c} \\ 0 & 0 \end{bmatrix}, \quad C = A S_c = \begin{bmatrix} W & 0 \\ A_{\mathcal{R}^c, \mathcal{C}} & 0 \end{bmatrix}.$$

Then analyses the error, and define $P_{\text{col}} = W W^+$ and $P_{\text{row}} = W^+ W$.

$$\begin{aligned} A - CUR &= \begin{bmatrix} 0 & (I - P_{\text{col}}) A_{\mathcal{R}, \mathcal{C}^c} \\ A_{\mathcal{R}^c, \mathcal{C}} (I - P_{\text{row}}) & A_{\mathcal{R}^c, \mathcal{C}^c} - A_{\mathcal{R}^c, \mathcal{C}} W^+ A_{\mathcal{R}, \mathcal{C}^c} \end{bmatrix} \\ \implies \|A - CUR\|_F &\leq \|(I - P_{\text{col}}) A_{\mathcal{R}, \mathcal{C}^c}\|_F + \|A_{\mathcal{R}^c, \mathcal{C}} (I - P_{\text{row}})\|_F + \|A_{\mathcal{R}^c, \mathcal{C}^c} - A_{\mathcal{R}^c, \mathcal{C}} W^+ A_{\mathcal{R}, \mathcal{C}^c}\|_F. \end{aligned} \quad (3)$$

If $\text{Column}(W) = \text{Column}(R)$, meaning that W preserves the column space of R . Then, because the column space of $R = [W \ A_{\mathcal{R}, \mathcal{C}^c}]$ is the sum of the column space of W and the column space of $A_{\mathcal{R}, \mathcal{C}^c}$, we have $\text{Column}(A_{\mathcal{R}, \mathcal{C}^c}) \subseteq \text{Column}(W)$. Therefore, the orthogonal projection onto the complement of the column space of W is zero, that is, $(I - P_{\text{col}}) A_{\mathcal{R}, \mathcal{C}^c} = 0$. Similarly, if W preserve the row space of C , that is $A_{\mathcal{R}^c, \mathcal{C}} (I - P_{\text{row}}) = 0$. All the assumption above will holds because of Lemma b with c and r is larger than $2500\theta^2$ with at least probability $(1 - \delta)^2 = 98\% * 0.98\% = 96.04\%$, δ is the failure probability of

Markov's inequality. Note that the row and column sampling probabilities p_i and q_j here are not exactly equivalent to the corresponding sampling probabilities in Lemma b. Taking row sampling as an example, p_i is computed based on the row ℓ_2 norms of A here, whereas the sampling probabilities in Lemma b are based on the row ℓ_2 norms of C . However, since C is obtained by randomly sampling columns from A , the two sampling distributions are approximately the same under random sampling, and this difference does not affect the application of Lemma b. Thus, Equation (3) can be converted as follows.

$$\begin{aligned}
\|A - CUR\|_F &= \left\| \begin{bmatrix} 0 & 0 \\ 0 & A_{\mathcal{R}^c, C^c} - A_{\mathcal{R}^c, C} W^+ A_{\mathcal{R}, C^c} \end{bmatrix} \right\|_F \\
&= \|A_{\mathcal{R}^c, C^c} - A_{\mathcal{R}^c, C} W^+ A_{\mathcal{R}, C^c}\|_F \\
&\leq \|A_{\mathcal{R}^c, C^c}\|_F + \|A_{\mathcal{R}^c, C}\|_F \|W^+\|_2 \|A_{\mathcal{R}, C^c}\|_F \\
&= \|A_{\mathcal{R}^c, C^c}\|_F + \frac{\|A_{\mathcal{R}^c, C}\|_F \|A_{\mathcal{R}, C^c}\|_F}{\sigma_{\min}(W)},
\end{aligned} \tag{4}$$

where $\|W^+\|_2 = \frac{1}{\sigma_{\min}(W)}$ and $\sigma_{\min}(W)$ is the minimum not zero singular value of W . Then, to further simplify this bound, we analyze $\|A_{\mathcal{R}^c, C^c}\|_F$, $\|A_{\mathcal{R}^c, C}\|_F$, and $\|A_{\mathcal{R}, C^c}\|_F$. We define two parameters α_c and α_r .

$$\alpha_c = \frac{\sum_j \|A_{:,j}\|_2^4}{\|A\|_F^4}, \quad \alpha_r = \frac{\sum_i \|A_{i,:}\|_2^4}{\|A\|_F^4}.$$

Then, we have

$$\begin{aligned}
\|A_{\mathcal{R}, C^c}\|_F^2 &= \sum_{i \in \mathcal{R}} \sum_{j \notin C} A_{ij}^2 \leq \sum_{i=1}^n \sum_{j \notin C} A_{ij}^2 = \|A_{:, C^c}\|_F^2 \\
\implies \mathbb{E}[\|A_{\mathcal{R}, C^c}\|_F^2] &\leq \mathbb{E}[\|A_{:, C^c}\|_F^2] = \sum_j \|A_{:,j}\|_2^2 (1 - cq_j) = \|A\|_F^2 - \frac{c}{\|A\|_F^2} \sum_j \|A_{:,j}\|_2^4 = (1 - c\alpha_c) \|A\|_F^2.
\end{aligned}$$

Similarly, we have

$$\|A_{\mathcal{R}^c, C}\|_F^2 \leq \|A_{\mathcal{R}^c, :}\|_F^2, \implies \mathbb{E}[\|A_{\mathcal{R}^c, C}\|_F^2] \leq (1 - r\alpha_r) \|A\|_F^2.$$

Then, we can also have

$$\begin{aligned}
\|A_{\mathcal{R}^c, C^c}\|_F^2 &= \sum_{i \notin \mathcal{R}} \sum_{j \notin C} A_{ij}^2 \leq \min \{ \|A_{:, C^c}\|_F^2, \|A_{\mathcal{R}^c, :}\|_F^2 \} \leq \sqrt{\|A_{:, C^c}\|_F^2 \cdot \|A_{\mathcal{R}^c, :}\|_F^2} = \|A_{:, C^c}\|_F \cdot \|A_{\mathcal{R}^c, :}\|_F \\
\implies \mathbb{E}[\|A_{\mathcal{R}^c, C^c}\|_F^2] &\leq \mathbb{E}[\|A_{:, C^c}\|_F \cdot \|A_{\mathcal{R}^c, :}\|_F] \leq \sqrt{\mathbb{E}[\|A_{:, C^c}\|_F^2] \cdot \mathbb{E}[\|A_{\mathcal{R}^c, :}\|_F^2]} \leq \sqrt{(1 - c\alpha_c)(1 - r\alpha_r)} \|A\|_F^2.
\end{aligned}$$

Here we clarify that p_i and q_j denote the sampling probabilities of the i -th row and the j -th column, respectively, rather than their inclusion probabilities, i.e., the actual probabilities that they are selected into the sample. When sampling r rows and c columns, the inclusion probabilities of the i -th row and the j -th column can be approximately expressed as rp_i and cq_j , respectively. This approximation is reasonable under the given sampling scheme. Moreover, since $c, r \ll n$, the assumptions $1 - rp_i > 0$ and $1 - cq_j > 0$ are easily satisfied.

Using Markov's inequality [4], convert the expectation bound obtained above into a probabilistic bound, and set $\delta = 0.02$ i.e., requiring the bound to hold with probability at least 98%.

$$\begin{aligned}
\|A_{\mathcal{R}^c, C}\|_F &\leq 5\sqrt{2}(1 - c\alpha_c)^{1/2} \|A\|_F \\
\|A_{\mathcal{R}, C^c}\|_F &\leq 5\sqrt{2}(1 - r\alpha_r)^{1/2} \|A\|_F \\
\|A_{\mathcal{R}^c, C^c}\|_F &\leq 5\sqrt{2}(1 - c\alpha_c)^{1/4} (1 - r\alpha_r)^{1/4} \|A\|_F
\end{aligned}$$

With probability at least $(1 - \delta)^3 > 94\%$, the error bound holds.

$$\begin{aligned}
\|A - CUR\|_F &\leq 5\sqrt{2}(1 - c\alpha_c)^{1/4} (1 - r\alpha_r)^{1/4} \|A\|_F + \frac{\left(5\sqrt{2}(1 - c\alpha_c)^{1/2} \|A\|_F\right) \left(5\sqrt{2}(1 - r\alpha_r)^{1/2} \|A\|_F\right)}{\sigma_{\min}(W)} \\
&= 5\sqrt{2}(1 - \rho n \alpha_c)^{1/4} (1 - \rho n \alpha_r)^{1/4} \|A\|_F + \frac{50(1 - \rho n \alpha_r)^{1/2} (1 - \rho n \alpha_c)^{1/2}}{\sigma_{\min}(W)} \|A\|_F^2.
\end{aligned} \tag{5}$$

Equation (2) has been provided in the previous work [5].

$$\|\pi - \tilde{\pi}\|_2 \leq \frac{\alpha}{1-\alpha} \cdot \sqrt{n} \cdot \|A - CUR\|_F \cdot \left\| \begin{bmatrix} 1 \\ n \end{bmatrix}_{n \times 1} \right\|_F = \frac{\alpha}{1-\alpha} \|A - CUR\|_F.$$

Lastly, α_r and α_c can be bounded as followed because of Cauchy-Schwarz inequality [3].

$$\alpha_r = \frac{\sum_i \|A_{i,:}\|_2^4}{\|A\|_F^4} \geq \frac{(\sum_i \|A_{i,:}\|_2^2)^2}{n \|A\|_F^4} = \frac{1}{n}, \quad \alpha_c = \frac{\sum_j \|A_{:,j}\|_2^4}{\|A\|_F^4} \geq \frac{(\sum_j \|A_{:,j}\|_2^2)^2}{n \|A\|_F^4} = \frac{1}{n}.$$

Then, the error bound can be further simplified.

$$\|A - CUR\|_F \leq 5\sqrt{2}(1-\rho)^{1/2} \|A\|_F + \frac{50(1-\rho)}{\sigma_{\min}(W)} \|A\|_F^2.$$

□

Noted: we discuss the Lemma 3.1 again. To guarantee with high probability that W preserves the row and column spaces of C and R , respectively, the quantities c and r are required to be very large in Lemma b. This is a stringent condition because the proof of the lemma uses sampling with replacement, whereas the actual algorithm employs sampling without replacement, so some irrelevant rows and columns will not be repeatedly sampled multiple times. Moreover, even if $\text{Column}(W) = \text{Column}(R)$ and $\text{Row}(W) = \text{Row}(C)$ do not hold exactly, the error bound in Lemma 3.1 retains the form of Equation(3), with only the additional terms $\|(I - P_{\text{col}})A_{\mathcal{R},C^c}\|_F$ and $\|A_{\mathcal{R}^c,C}(I - P_{\text{row}})\|_F$. If $\text{Rank}(W) = k$, we have

$$\begin{aligned} \|(I - P_{\text{col}})A_{\mathcal{R},C^c}\|_F &\leq \sqrt{r-k} \cdot \|A_{\mathcal{R},C^c}\|_2 \leq \sqrt{r-k} \\ \|A_{\mathcal{R}^c,C}(I - P_{\text{row}})\|_F &\leq \sqrt{c-k} \cdot \|A_{\mathcal{R}^c,C}\|_2 \leq \sqrt{c-k}. \end{aligned}$$

Lemma. 3.1.a. Given $\theta \leq \text{rank}(A)$, let A_θ denote the optimal rank- θ approximation of A . If the column and row sampling ratio in CUR-Trans satisfies $\rho > \frac{\theta^2}{n\delta^2}$, then the following bound holds with probability at least 0.9.

$$\mathbb{E} [\|A - CUR\|_F] \leq (1 - \rho\alpha_c)^{1/4} (1 - \rho\alpha_r)^{1/4} \|A\|_F + \frac{(1 - \rho\alpha_r)^{1/2} (1 - \rho\alpha_c)^{1/2}}{\sigma_{\min}(W)} \|A\|_F^2.$$

where W is the intersection of matrices C and R , $\sigma_{\min}(W)$ the smallest non-zero singular value W , $\alpha_r = \sum_i \|A_{i,:}\|_2^4 / \|A\|_F^4 \geq 1/n$, and $\alpha_c = \sum_j \|A_{:,j}\|_2^4 / \|A\|_F^4 \geq 1/n$.

Proof. Keep the expectation form, we have

$$\mathbb{E} [\|A - CUR\|_F] \leq (1 - \rho\alpha_c)^{1/4} (1 - \rho\alpha_r)^{1/4} \|A\|_F + \frac{(1 - \rho\alpha_r)^{1/2} (1 - \rho\alpha_c)^{1/2}}{\sigma_{\min}(W)} \|A\|_F^2. \quad (6)$$

Apply the Markov's inequality, With the probability at least $(1 - \delta)^5$, we have

$$\|A - CUR\|_F \leq \frac{1}{\sqrt{\delta}} (1 - \rho\alpha_c)^{1/4} (1 - \rho\alpha_r)^{1/4} \|A\|_F + \frac{1}{\delta} \cdot \frac{(1 - \rho\alpha_r)^{1/2} (1 - \rho\alpha_c)^{1/2}}{\sigma_{\min}(W)} \|A\|_F^2. \quad (7)$$

□

Lemma. 3.1.b. W is a matrix obtained by sampling r rows from C according to probability p_i , meaning that the i -th rows is sampled from C with probability p_i , and $\varrho_c = \min_i \left\{ p_i \cdot \frac{\|C\|_F^2}{\|C_{i,:}\|_2^2} \right\}$. If $\text{Rank}(C) = \mu$ and $\theta \geq \mu$, for $r > \frac{\theta^2}{\delta^2 \varrho_c}$, then with a probability at least $1 - \delta$, W can preserve the row subspace of C , i.e., $\text{Row}(C) = \text{Row}(W)$.

W is a matrix obtained by sampling c columns from R according to probability q_j , meaning that the j column is sampled from R with probability q_j , and $\varrho_r = \min_j \left\{ p_j \cdot \frac{\|R\|_F^2}{\|R_{:,j}\|_2^2} \right\}$. If $\text{Rank}(R) = \mu$ and $\theta \geq \mu$, for $c > \frac{\theta^2}{\delta^2 \varrho_r}$, then with a probability at least $1 - \delta$, W can preserve the column subspace of R , i.e., $\text{Column}(R) = \text{Column}(W)$.

Proof. Because $\text{Rank}(C) = \mu$, the SVD decomposition of C is $C_{n \times c} = U_{C,\mu} \Sigma_{C,\mu} V_{C,\mu}^\top$. D is a diagonal matrix denoted the scaling of sampled rows, $D_{i,i} = \frac{1}{\sqrt{r p_i}}$, else $D_{i,i} = 0$. Let $\mathbb{S} = D \cdot S_r$. We have

$$|1 - \sigma_i^2(\mathbb{S} U_{C,\theta})| = |\sigma_i(U_{C,\theta}^\top U_{C,\theta}) - \sigma_i(U_{C,\theta}^\top \mathbb{S}^\top \mathbb{S} U_{C,\theta})| \quad (8)$$

$$\leq \|U_{C,\theta}^\top U_{C,\theta} - U_{C,\theta}^\top \mathbb{S}^\top \mathbb{S} U_{C,\theta}\|_2 \quad (9)$$

$$\leq \frac{1}{\delta} \mathbb{E} \left[\|U_{C,\theta}^\top U_{C,\theta} - U_{C,\theta}^\top \mathbb{S}^\top \mathbb{S} U_{C,\theta}\|_2 \right] \quad (10)$$

$$\leq \frac{1}{\delta} \sqrt{\frac{1}{r \varrho_c}} \|U_{C,\theta}\|_F^2 \quad (11)$$

$$= \frac{1}{\delta} \sqrt{\frac{1}{r \varrho_c}} \theta \quad (12)$$

$$< 1. \quad (13)$$

Equation (8) holds from properties of SVD that $U_{C,\theta}^\top U_{C,\theta} = I$ and $\sigma_i^2(\mathbb{S} U_{C,\theta}) = \sigma_i(U_{C,\theta}^\top \mathbb{S}^\top \mathbb{S} U_{C,\theta})$. Equation (9) holds from $|\sigma(C + E) - \sigma(C)| \leq \|E\|_2$ [2]. Equation (10) holds from the Markov's inequity [4], which is that given $\delta > 0$, $P(X \geq \delta) \leq \frac{E(X)}{\delta}$. Then, we give $a = \frac{1}{\delta} \mathbb{E} \left[\|U_{C,\theta}^\top U_{C,\theta} - U_{C,\theta}^\top \mathbb{S}^\top \mathbb{S} U_{C,\theta}\|_2 \right]$, so Equation (10) holds with probability at least $1 - \delta$. Equation (11) holds because of Theorem 7 in work [1]. Equation (12) holds because $\|U_{C,\theta}\|_F^2 = \|U_{C,\theta}^\top U_{C,\theta}\|_F^2 = \theta$. Equation (13) holds because $r > \frac{\theta^2}{\delta^2 \varrho_c}$. Therefore, all singular values of $\mathbb{S} U_{C,\theta}$ are strictly positive, which means that $\text{Rank}(\mathbb{S} U_{C,\theta}) = \text{Rank}(U_{C,\theta})$. Because D is a linear transformation matrix for $\mathbb{S} U_{C,\theta}$, which will not change the rank, so with a probability at least 98%, we have $\text{Rank}(\mathbb{S} U_{C,\theta}) = \text{Rank}(U_{C,\theta}) = \text{Rank}(C_\theta)$. Therefore, the matrix W preserve the row subspace of matrix C , i.e., $\text{Row}(C) = \text{Row}(W)$. The proof is the same for the matrix W when it is constructed by sampling columns from R , and we only need to replace the matrix $U_{C,\theta}$ with $V_{R,\theta}^\top$. $\square$

2 Lemma 3.2 Sequential T^2 -Approx

Lemma. 3.2 Let ρ denote the column and row sampling ratio in T^2 -Approx. The following bound holds with probability at least 0.9.

$$\|A^2 - XY\|_F \leq \sqrt{10(1 - \rho n \alpha_{cr})} \|A\|_F^2, \quad (14)$$

where $\alpha_{cr} = (\sum_{i=1}^n \|A_{:,i}\|_2^3 \|A_{i,:}\|_2) / (\|A\|_F^2 \cdot \sum_{k=1}^n \|A_{:,k}\|_2 \|A_{k,:}\|_2) \geq \bar{d} / n d_{\max}$, $\bar{d}$ is the average degree of graph, and $d_{\max}$ is the maximum degree of graph G . Consequently, the PageRank approximation error of T^2 -Approx satisfies

$$\|\pi - \tilde{\pi}\|_2 \leq \frac{\alpha}{2n(1 - \alpha)} \|A^2 - XY\|_F. \quad (15)$$

Proof. The notation follows the proof of Lemma 3.1. Since the rows and columns are sampled synchronously, i.e., $S_r = S_c = S$, we have $C = AS$ and $R = SA$.

$$\begin{aligned} CR &= ASAS = ASA \\ \implies A^2 - CR &= A^2 - ASA = A(I - S)A \\ \implies \|A^2 - CR\|_F &= \|A(I - S)A\|_F \leq \|A(I - S)\|_F \|A\|_2 \leq \|A(I - S)\|_F \|A\|_F. \end{aligned}$$

The sampling probabilities of vertex i being included as a column in C and as a row in R are given by:

$$p_i = \frac{\|A_{:,i}\|_2 \|A_{i,:}\|_2}{\sum_{k=1}^n \|A_{:,k}\|_2 \|A_{k,:}\|_2}.$$

Since $I - S$ only retains the columns corresponding to the unsampled vertices, and the rows and columns are sampled synchronously, let $\Omega = \mathcal{R} = \mathcal{C}$ denote the index set of the sampled rows and columns. Then, define a parameter α_{cr} .

$$\alpha_{cr} = \frac{\sum_{i=1}^n \|A_{:,i}\|_2^3 \|A_{i,:}\|_2}{(\sum_{k=1}^n \|A_{:,k}\|_2 \|A_{k,:}\|_2) \|A\|_F^2}.$$

Then, we have

$$\mathbb{E} [\|A(I - S)\|_F^2] = \mathbb{E} \left[\sum_{i \notin \Omega} \|A_{:,i}\|_2^2 \right] = \|A\|_F^2 - \frac{c}{\sum_{k=1}^n \|A_{:,k}\|_2 \|A_{k,:}\|_2} \sum_{i=1}^n \|A_{:,i}\|_2^3 \|A_{i,:}\|_2 = (1 - c\alpha_{cr}) \|A\|_F^2.$$

Using Markov's inequality, and set $\delta = 0.1$ i.e., requiring the bound to hold with probability at least 90%.

$$\|A(I - S)\|_F \leq \sqrt{10}(1 - c\alpha_{cr})^{1/2} \|A\|_F^2.$$

Because $C = X$ and $R = Y$ in the T^2 -Approx, the following inequality holds with probability at least 90%.

$$\|A^2 - XY\|_F \leq \sqrt{10}(1 - c\alpha_{cr})^{1/2} \|A\|_F^2 \quad (16)$$

Equation (15) has been provided in the previous work [5].

$$\|\pi - \tilde{\pi}\|_2 \leq \frac{\alpha}{2\sqrt{n}(1 - \alpha)} \|A^2 - XY\|_F \left\| \begin{bmatrix} 1 \\ n \end{bmatrix}_{n \times 1} \right\| = \frac{\alpha}{2n(1 - \alpha)} \|A^2 - XY\|_F.$$

Last, compute the bound of α_{rd} . Since $\|A_{:,i}\|_2^2 \geq \min_i \|A_{:,i}\|_2^2$, we have

$$\sum_i \|A_{i,:}\|_2 \|A_{:,i}\|_2^3 = \sum_i \|A_{i,:}\|_2 \|A_{:,i}\|_2 \|A_{:,i}\|_2^2 \geq \min_i \|A_{:,i}\|_2^2 \sum_i \|A_{i,:}\|_2 \|A_{:,i}\|_2 = \frac{1}{d_{\max}} \sum_i \|A_{i,:}\|_2 \|A_{:,i}\|_2.$$

Using the Cauchy-Schwarz inequality [3], we have

$$\|A\|_F^2 = \sum_i \frac{1}{d_i} \geq \frac{n^2}{\sum_i d_i} = \frac{n}{\bar{d}}.$$

Therefore, we have

$$\alpha_{rd} \geq \frac{\min_i \|A_{:,i}\|_2^2}{\|A\|_F^2} \geq \frac{1/d_{\max}}{\sum_i 1/d_i} \geq \frac{\bar{d}}{nd_{\max}}.$$

Then, the error bound can be further simplified.

$$\|A^2 - XY\|_F \leq \sqrt{10} \left(1 - \frac{c\bar{d}}{nd_{\max}} \right)^{1/2} \|A\|_F^2.$$

Equation (??) has been proved in the previous work [5], so the lemma is proved. $\square$

Lemma. (c) Given $\theta \leq \text{rank}(A)$, let A_θ denote the optimal rank- θ approximation of A . If the column and row sampling ratio in CUR-Trans satisfies $\rho > \frac{\theta^2}{n\delta^2}$, then the following bound holds with probability at least 0.9.

$$\mathbb{E} [\|A^2 - XY\|_F] \leq (1 - \rho\alpha_c)^{1/4} (1 - \rho\alpha_r)^{1/4} \|A\|_F + \frac{(1 - \rho\alpha_r)^{1/2} (1 - \rho\alpha_c)^{1/2}}{\sigma_{\min}(W)} \|A\|_F^2.$$

where W is the intersection of matrices C and R , $\sigma_{\min}(W)$ the smallest non-zero singular value W , $\alpha_r = \sum_i \|A_{i,:}\|_2^4 / \|A\|_F^4 \geq 1/n$, and $\alpha_c = \sum_j \|A_{:,j}\|_2^4 / \|A\|_F^4 \geq 1/n$.

Proof. Keep the expectation form, we have

$$\mathbb{E} [\|A^2 - CR\|_F] \leq \mathbb{E} [\|A(I - S)\|_F] \mathbb{E} [\|A\|_F] = \sqrt{(1 - c\alpha_{cr})} \|A\|_F^2. \quad (17)$$

Apply the Markov's inequality, With the probability at least $1 - \delta$, we have

$$\|A - CUR\|_F \leq \frac{1}{\delta} \sqrt{(1 - c\alpha_{cr})} \|A\|_F^2. \quad (18)$$

$\square$

3 Lemma 4.1 Parallel CUR-Trans

Lemma. 4.1 Given a sampling ratio $\rho > \frac{2500\theta^2}{nD}$, the error bound of the MSSP framework using CUR-Trans satisfies

$$\mathcal{E}_{CUR}^{(D)} \leq \frac{D - \rho}{D - D\rho} \cdot \mathcal{E}_{CUR}.$$

Proof. Define $H(\rho) = \sqrt{50}(1 - \rho n\alpha_c)^{1/4}(1 - \rho n\alpha_r)^{1/4}\|A\|_F$, $F(\rho) = \frac{50(1 - \rho n\alpha_r)^{1/2}(1 - \rho n\alpha_c)^{1/2}}{\sigma_{\min}(W)}\|A\|_F^2$, and $f(\rho) = H(\rho) + F(\rho)$.

Let $a = n\alpha_c$, $b = n\alpha_r$, and we have $0 < \rho < 1/\max(a, b)$. Define $x(\rho) = \sqrt{(1 - a\rho)(1 - b\rho)}$, then we have

$$\begin{aligned} H(\rho) &= \sqrt{50}\|A\|_F(1 - \rho n\alpha_c)^{1/4}(1 - \rho n\alpha_r)^{1/4} = \sqrt{50}\|A\|_F\sqrt{x(\rho)}, \\ F(\rho) &= \frac{50(1 - \rho n\alpha_c)^{1/2}(1 - \rho n\alpha_r)^{1/2}}{\sigma_{\min}(W)}\|A\|_F^2 = \frac{50\|A\|_F^2}{\sigma_{\min}(W)}x(\rho). \end{aligned}$$

Then, define that $h = \sqrt{50}\|A\|_F$ and $g = \frac{50\|A\|_F^2}{\sigma_{\min}(W)}$, and we have

$$f(\rho) = h\sqrt{x(\rho)} + gx(\rho).$$

For ρ/D , we have

$$x(\rho/D) = \sqrt{\left(1 - \frac{a\rho}{D}\right)\left(1 - \frac{b\rho}{D}\right)}.$$

Since $D > 1$, it follows that

$$x(\rho/D) \geq x(\rho).$$

Let $t = \frac{x(\rho/D)}{x(\rho)} \geq 1$, then

$$\frac{f(\rho/D)}{f(\rho)} = \frac{h\sqrt{x(\rho/D)} + gx(\rho/D)}{h\sqrt{x(\rho)} + gx(\rho)} = \frac{h\sqrt{x(\rho)}\sqrt{t} + gx(\rho)t}{h\sqrt{x(\rho)} + gx(\rho)} = \frac{h\sqrt{x(\rho)}}{h\sqrt{x(\rho)} + gx(\rho)}\sqrt{t} + \frac{gx(\rho)}{h\sqrt{x(\rho)} + gx(\rho)}t.$$

The two coefficients are nonnegative and sum to one. Therefore, the above expression is a convex combination of $\sqrt{t}$ and t . Since $t \geq \sqrt{t}$, we obtain

$$\frac{f(\rho/D)}{f(\rho)} \leq t.$$

Hence, we have

$$\frac{f(\rho/D)}{f(\rho)} \leq \frac{x(\rho/D)}{x(\rho)} \implies \frac{f(\rho/D)}{f(\rho)} \leq \sqrt{\frac{(1 - \frac{\rho n\alpha_c}{D})(1 - \frac{\rho n\alpha_r}{D})}{(1 - \rho n\alpha_c)(1 - \rho n\alpha_r)}}.$$

Furthermore, because $\alpha_c \geq \frac{1}{n}$, and $\alpha_r \geq \frac{1}{n}$, the weakest case is $n\alpha_c = n\alpha_r = 1$. Therefore, we have

$$\frac{f(\rho/D)}{f(\rho)} \leq \frac{1 - \rho/D}{1 - \rho} = \frac{D - \rho}{D - D\rho}.$$

Define the upper of the error bound of each sample is $E_{CUR}^{(D)}$, and we have

$$\|\pi - \tilde{\pi}_d\|_2 \leq E_{CUR}^{(D)} = \frac{\alpha}{1 - \alpha} \cdot f(D/c).$$

Moreover, if W changes with the sampling ratio ρ , then the above bound should additionally include the variation of the minimum singular value $\sigma_{\min}(W(\rho))/\sigma_{\min}(W(\rho/D))$. In this case, the ratio cannot be bounded only by α_c, α_r, ρ , and D without an additional lower bound on $\sigma_{\min}(W)$. However, $\rho > \frac{2500\theta^2}{nD}$ is required to guarantee the subspace coverage of W , which means that $\sigma_{\min}(W)$ is assumed to remain unchanged.

According to Lemma 3.1, $\mathcal{E}_{CUR} = \frac{\alpha}{1 - \alpha} \cdot f(\rho)$. Therefore, because of uniform weighted, we have

$$\begin{aligned} \|\pi - \tilde{\pi}^{(D)}\|_2 &= \left\| \pi - \sum_{d=1}^D \omega_d \cdot \tilde{\pi}_d \right\|_2 \leq \sum_{d=1}^D \omega_d \|\pi - \tilde{\pi}_d\|_2 = 1 \cdot \frac{D - \rho}{D - D\rho} \cdot \mathcal{E}_{CUR} \\ \implies \mathcal{E}_{CUR}^{(D)} &\leq \frac{D - \rho}{D - D\rho} \mathcal{E}_{CUR}. \end{aligned}$$

□

4 Lemma 4.2 Parallel T^2 -Approx

Lemma. *Given a sampling ratio $\rho > 0$, the error bound of the MSSP framework using T^2 -Approx satisfies*

$$\mathcal{E}_{T^2}^{(D)} \leq \sqrt{\frac{Dd_{\max} - \bar{d}\rho}{Dd_{\max} - D\bar{d}\rho}} \cdot \mathcal{E}_{T^2},$$

where $\bar{d}$ is the average degree of the graph G and $d_{\max}$ is the maximum degree of the graph.

Proof. The proof is the same as Lemma 4.1. Define $f(c) = \sqrt{10} \left(1 - \frac{c\bar{d}}{nd_{\max}}\right)^{1/2} \|A\|_F^2$.

$$\frac{f(c/D)}{f(c)} = \sqrt{\frac{Dd_{\max} - \bar{d}\rho}{Dd_{\max} - D\bar{d}\rho}},$$

and $f(c)$ is the error bound of $\|A^2 - XY\|_F$ when the number of sampled column and row is c . According to Lemma 3.2, the error bound for $|\pi - \tilde{\pi}|_2$ scales linearly with the error bound for $\|A^2 - XY\|_F$, and it can be denoted that $\mathcal{E}_{T^2} = \mathcal{K} \cdot f(c)$, and $\mathcal{K}$ is a constant.

$$\begin{aligned} \|\pi - \tilde{\pi}^{(D)}\|_2 &\leq \sum_{d=1}^D \omega_d \|\pi - \tilde{\pi}_d\|_2 \leq \sum_{d=1}^D \omega_d \cdot \mathcal{K} \cdot f(c/D) \leq \sum_{d=1}^D \omega_d \cdot \mathcal{K} \cdot \sqrt{\frac{Dd_{\max} - \bar{d}\rho}{Dd_{\max} - D\bar{d}\rho}} \cdot f(c) = \sqrt{\frac{Dd_{\max} - \bar{d}\rho}{Dd_{\max} - D\bar{d}\rho}} \cdot \mathcal{E}_{T^2} \\ &\Rightarrow \frac{\mathcal{E}_{T^2}^{(D)}}{\mathcal{E}_{T^2}} \leq \sqrt{\frac{Dd_{\max} - \bar{d}\rho}{Dd_{\max} - D\bar{d}\rho}}. \end{aligned}$$

□

5 Lemma 4.3 Parallel T^2 -Approx with Shuffled Partitioning

Lemma. 4.4 *If $\rho > 0$, then, with probability at least 0.86, the error-bound adjustment induced by shuffled partitioning for T^2 -Approx is*

$$\Delta \mathcal{E}_{T^2-P}^{(D)} = 20 \sqrt{\left(1 - \frac{1}{D}\right) \frac{\alpha \|A\|_F}{2n(1-\alpha)}} + 10 \sqrt{(1 - N\alpha_{cr})} \frac{\alpha \|A\|_F^2}{2nD(1-\alpha)}.$$

Proof. All parameter definitions and proof steps follow the proof of Lemma 4.5. We have

$$\|A^2 - XY\|_F \leq \|A^2 - A_d^2\|_F + \|A_d^2 - XY\|_F.$$

First, we consider the error introduced by the shuffled partitioning. Because $A^2 - A_d^2 = (A - A_d)A + A_d(A - A_d)$, then

$$\|A^2 - A_d^2\|_F \leq \|A - A_d\|_F \|A\|_2 + \|A_d\|_2 \|A - A_d\|_F \leq (\|A\|_2 + \|A_d\|_2) \|A - A_d\|_F \leq 2\|A\|_2 \|A - A_d\|_F \leq 2\|A - A_d\|_F.$$

Then, we have that

$$\begin{aligned} \mathbb{E}[\|A - A_d\|_F^2] &= \sum_i \sum_j \left(1 - \frac{N}{n}\right) A_{i,j}^2 = \left(1 - \frac{1}{D}\right) \|A\|_F^2 \\ \mathbb{E}[\|A_d\|_F^2] &= \frac{1}{D} \|A\|_F^2. \end{aligned}$$

Let $N = \rho n/D$.

$$\begin{aligned} \mathbb{E}[\|A^2 - XY\|_F] &\leq \mathbb{E}[\|A^2 - A_d^2\|_F] + \mathbb{E}[\|A_d^2 - XY\|_F] \\ &\leq 2\mathbb{E}[\|A - A_d\|_F] + \sqrt{(1 - N\alpha_{cr})} \mathbb{E}[\|A_d\|_F^2] \\ &\leq 2\sqrt{\left(1 - \frac{1}{D}\right)} \|A\|_F + \sqrt{(1 - N\alpha_{cr})} \frac{1}{D} \|A\|_F^2. \end{aligned}$$

Define the upper of the error bound of each sample is $E_{T^2}^{(D)}$ without shuffled partitioning, $\psi = (1 - N\alpha_c)^{1/4}(1 - N\alpha_r)^{1/4}$,

$$\begin{aligned}\mathbb{E} [\|A^2 - XY\|_F] - \mathbb{E} [E_{T^2}^{(D)}] &\leq 2\sqrt{\left(1 - \frac{1}{D}\right)}\|A\|_F + \sqrt{\frac{1}{D}}\sqrt{(1 - N\alpha_{cr})}\|A\|_F^2 - \sqrt{(1 - N\alpha_{cr})}\|A\|_F^2 \\ &= 2\sqrt{\left(1 - \frac{1}{D}\right)}\|A\|_F + \left(\frac{1}{\sqrt{D}} - 1\right)\sqrt{(1 - N\alpha_{cr})}\|A\|_F^2\end{aligned}$$

Let $x = \sqrt{D}$, $x > 1$

$$\begin{aligned}g(D) &= 2\sqrt{1 - \frac{1}{D}}\|A\|_F + \left(\frac{1}{\sqrt{D}} - 1\right)\sqrt{1 - N\alpha_{cr}}\|A\|_F^2 \\ g(x) &= 2\frac{\sqrt{x^2 - 1}}{x} - \frac{x - 1}{x}\sqrt{1 - N\alpha_{cr}}\|A\|_F.\end{aligned}$$

We analyze the condition

$$\begin{aligned}g(x) &< 0 \\ 2\sqrt{x^2 - 1} &< (x - 1)\sqrt{1 - N\alpha_{cr}}\|A\|_F \\ 2\sqrt{(x - 1)(x + 1)} &< (x - 1)\sqrt{1 - N\alpha_{cr}}\|A\|_F \\ 2\sqrt{x + 1} &< \sqrt{x - 1}\sqrt{1 - N\alpha_{cr}}\|A\|_F \\ 4(x + 1) &< (x - 1)(1 - N\alpha_{cr})\|A\|_F^2 \\ 4(\sqrt{D} + 1) &< (1 - N\alpha_{cr})\|A\|_F^2 \\ \frac{\sqrt{D} + 1}{\sqrt{D} - 1} &< \frac{(1 - N\alpha_{cr})\|A\|_F^2}{4} \\ D &> \left(1 + \frac{8}{(1 - N\alpha_{cr})\|A\|_F^2 - 4}\right)^2.\end{aligned}$$

We also require $(1 - N\alpha_{cr})\|A\|_F^2 > 4$.

By Markov's inequality, with probability at least 98%, the following bound has been proved in the proof of the Lemma 4.5.

$$\|A - A_d\|_F \leq \sqrt{50}\sqrt{1 - \frac{1}{D}}\|A\|_F.$$

The sampling probabilities of vertex i being included as a column in C and as a row in R are given by:

$$p_i = \frac{\|A_{:,i}\|_2\|A_{i,:}\|_2}{\sum_{k=1}^n \|A_{:,k}\|_2\|A_{k,:}\|_2}.$$

From Lemma 3.2, we have

$$\|A_d^2 - XY\|_F \leq \sqrt{10}\left(1 - \frac{c\bar{d}}{nd_{\max}}\right)^{1/2}\|A_d\|_F^2 \leq \sqrt{10}\left(1 - \frac{c\bar{d}}{nd_{\max}}\right)^{1/2}\|A\|_F^2.$$

Consider the case without using shuffled partitioning with the sampled columns and rows is c/D for each sample, then we have

$$\|A^2 - XY\|_F \leq \sqrt{10}\left(1 - \frac{c\bar{d}}{Dnd_{\max}}\right)^{1/2}\|A\|_F^2.$$

The sampling probabilities of vertex i being included as a column in C and as a row in R are given by:

$$p_i = \frac{d_i\|A_{i,:}\|_2}{\sum_{k=1}^n d_k\|A_{:,k}\|_2}.$$

From Lemma 4.4, we have

$$\|A_d^2 - XY\|_F \leq \sqrt{10} \left(1 - \frac{cd_{\min}}{nd_{\max}}\right)^{1/2} \|A\|_F^2.$$

Consider the case without using shuffled partitioning with the sampled columns and rows is c/D for each sample, then we have

$$\|A_d^2 - XY\|_F \leq \sqrt{10} \left(1 - \frac{cd_{\min}}{Dnd_{\max}}\right)^{1/2} \|A\|_F^2.$$

The sample constructed without shuffled partitioning incurs a larger error bound than that constructed with shuffled partitioning. Therefore, compared with the case without shuffled partitioning, the quality of each individual sample under shuffled partitioning depends on the partition accuracy of each sample.

By Markov's inequality (with $\delta = 0.04$), with probability at least 0.98, the following bound holds.

$$\|A_d\|_F \leq \sqrt{\frac{25}{D}} \|A\|_F.$$

$$\|A^2 - XY\|_F \leq \sqrt{10} \left(1 - \frac{\rho n \alpha_{cr}}{D}\right) \|A\|_F^2$$

$$\sqrt{10} \left(1 - \frac{\rho n \alpha_{cr}}{D}\right) \frac{50}{D} \|A\|_F + 10 \sqrt{1 - \frac{1}{D}} \|A\|_F - \sqrt{10} \left(1 - \frac{\rho n \alpha_{cr}}{D}\right) \|A\|_F$$

□

Noted: The overall error decreases only when the error introduced by the shuffled partition, $\|A^2 - A_d^2\|_F$, is smaller than the gain achieved by sampling.

6 Lemma 4.4 Parallel CUR-Trans with Shuffled Partitioning

Lemma. 4.3 Given a sampling ratio $\rho > \frac{2500\theta^2}{nD}$, then, with probability at least 0.9, the error-bound adjustment induced by shuffled partitioning for CUR-Trans is

$$\Delta \mathcal{E}_{CUR-P}^{(D)} = \left(\frac{1 - \sqrt{D}}{\sqrt{D}} \psi + \frac{\sqrt{D-1}}{\sqrt{D}} \right) \frac{50\alpha \|A\|_F}{1 - \alpha} - \frac{(D-1)50\alpha\psi^2 \|A\|_F^2}{D \cdot \sigma_{\min}(W)(1 - \alpha)},$$

$$\text{where } \psi = (1 - N\alpha_c)^{1/4} (1 - N\alpha_r)^{1/4}.$$

Proof. The Shuffled partitioning operation, for each sample, is equivalent to first uniformly sampling N ($N = \frac{n}{D}$) rows and N columns from A to form A_d , and then sampling c/D ($< N$) columns and r/D ($< N$) rows from A_d to form C and R . Let $\tilde{A}_d$ denote the CUR approximation obtained after shuffled partitioning, where $\tilde{A}_d$ samples c/D ($< N$) columns and r/D ($< N$) rows from A_d . Other notations are the same as previous lemma proofs. Define the upper of the error bound of each sample with shuffled partitioning is $E_{CUR-S}^{(D)}$, and $\|A - \tilde{A}_d\|_F \leq E_{CUR-S}^{(D)}$.

$$\|A - \tilde{A}_d\|_F \leq \|A - A_d\|_F + \|A_d - \tilde{A}_d\|_F.$$

The quality of a single sample constructed via shuffled partitioning consists of two components: one is the partitioning error $\|A - A_d\|_F$, and the other is the construction error of the sample $\|A_d - \tilde{A}_d\|_F$ (i.e., the construction of each iterative matrix in CUR-Trans). In contrast, the quality of a single sample constructed without shuffled partitioning consists solely of the construction error.

First, calculate the bound of $\|A - A_d\|_F$ and $\|A_d\|_F$.

$$\mathbb{E} [\|A - A_d\|_F^2] = \sum_i \sum_j \left(1 - \frac{N}{n}\right) A_{i,j}^2 = \left(1 - \frac{1}{D}\right) \|A\|_F^2$$

$$\mathbb{E} [\|A_d\|_F^2] = \frac{1}{D} \|A\|_F^2.$$

Let $N = \rho n/D$, then we have

$$\begin{aligned}\mathbb{E} [\|A - \tilde{A}_d\|_F] &\leq \mathbb{E} [\|A - \tilde{A}_d\|_F] + \mathbb{E} [\|A_d - \tilde{A}_d\|_F] \\ &\leq [(1 - N\alpha_c)^{1/4}(1 - N\alpha_r)^{1/4} + (D - 1)^{1/2}] \cdot \frac{\|A\|_F}{\sqrt{D}} + [(1 - N\alpha_r)^{1/2}(1 - N\alpha_c)^{1/2}] \cdot \frac{\|A\|_F^2}{D \cdot \sigma_{\min}(W)}\end{aligned}$$

Define the upper of the error bound of each sample is $E_{CUR}^{(D)}$ without shuffled partitioning, $\psi = (1 - N\alpha_c)^{1/4}(1 - N\alpha_r)^{1/4}$,

$$\begin{aligned}\mathbb{E} [\|A - \tilde{A}_d\|_F] - \mathbb{E} [E_{CUR}^{(D)}] &= [(1 - N\alpha_c)^{1/4}(1 - N\alpha_r)^{1/4} + (D - 1)^{1/2}] \cdot \frac{\|A\|_F}{\sqrt{D}} + [(1 - N\alpha_r)^{1/2}(1 - N\alpha_c)^{1/2}] \cdot \frac{\|A\|_F^2}{D \cdot \sigma_{\min}(W)} \\ &\quad - (1 - N\alpha_c)^{1/4}(1 - N\alpha_r)^{1/4}\|A\|_F - \frac{(1 - N\alpha_r)^{1/2}(1 - N\alpha_c)^{1/2}}{\sigma_{\min}(W)}\|A\|_F^2 \\ &= \left[(1 - N\alpha_c)^{1/4}(1 - N\alpha_r)^{1/4} \left(\frac{1 - D^{1/2}}{D^{1/2}} \right) + \left(\frac{D - 1}{D} \right)^{\frac{1}{2}} \right] \cdot \|A\|_F + \left[(1 - N\alpha_r)^{1/2}(1 - N\alpha_c)^{1/2} \left(\frac{1 - D}{D} \right) \right] \cdot \frac{\|A\|_F^2}{\sigma_{\min}(W)} \\ &= \left(\frac{1 - \sqrt{D}}{\sqrt{D}}\psi + \frac{\sqrt{D - 1}}{\sqrt{D}} \right) \|A\|_F + \frac{1 - D}{D} \cdot \frac{\psi^2\|A\|_F^2}{\sigma_{\min}(W)}.\end{aligned}$$

Let $\psi = (1 - N\alpha_c)^{1/4}(1 - N\alpha_r)^{1/4}$ and $x = \sqrt{D}$, and we have

$$\begin{aligned}g(D) &= \left(\frac{1 - \sqrt{D}}{\sqrt{D}}\psi + \frac{D - 1}{\sqrt{D}} \right) \|A\|_F + \frac{1 - D}{D} \frac{\psi^2\|A\|_F^2}{\sigma_{\min}(W)} \\ g(x) &= \left(\frac{1 - x}{x}\psi + \frac{\sqrt{x^2 - 1}}{x} \right) \|A\|_F - \frac{x^2 - 1}{x^2} \frac{\psi^2\|A\|_F^2}{\sigma_{\min}(W)}.\end{aligned}$$

Since $0 \leq \psi \leq 1$, and $\sqrt{x^2 - 1} > x - 1$, we have $\sqrt{x^2 - 1} - (x - 1)\psi > \sqrt{x^2 - 1} - (x - 1) > 0$. Therefore, the first term of $g(D)$ is always positive, while the second term is negative. The sign of $g(D)$ depends on the relative magnitude of these two terms.

$$\begin{aligned}g(D) &< 0 \\ \implies \sigma_{\min}(W) &> \frac{x(x^2 - 1)\psi^2\|A\|_F}{\sqrt{x^2 - 1} - (x - 1)\psi} \\ \implies \sigma_{\min}(W) &> \frac{\sqrt{D}(D - 1)\psi^2\|A\|_F}{\sqrt{D - 1} - (\sqrt{D} - 1)\psi}.\end{aligned}$$

Therefore, if $\sigma_{\min}(W) > \frac{\sqrt{D}(D - 1)\psi^2\|A\|_F}{\sqrt{D - 1} - (\sqrt{D} - 1)\psi}$, $g(D) < 0$. By Markov's inequality (with $\delta = 0.02$), with probability at least $0.96 > 0.98^3$, the following bound holds.

$$\|A - \tilde{A}_d\|_F - E_{CUR}^{(D)} \leq 50 \left[\left(\frac{1 - \sqrt{D}}{\sqrt{D}}\psi + \frac{\sqrt{D - 1}}{\sqrt{D}} \right) \|A\|_F - \frac{D - 1}{D} \cdot \frac{\psi^2\|A\|_F^2}{\sigma_{\min}(W)} \right].$$

The same as the proof of Lemma 4.1, because of uniform weighted, we have that the error change of a single sample is equivalent to the error change after aggregating all samples. Therefore, we have

$$\Delta\mathcal{E}_{CUR-P}^{(D)} = 50 \left[\left(\frac{1 - \sqrt{D}}{\sqrt{D}}\psi + \frac{\sqrt{D - 1}}{\sqrt{D}} \right) \|A\|_F - \frac{D - 1}{D} \cdot \frac{\psi^2\|A\|_F^2}{\sigma_{\min}(W)} \right]$$

□

Noted: The overall error decreases only when the error introduced by the shuffled partition, $\|A - A_d\|_F$, is smaller than the gain achieved by sampling.

7 Lemma 4.5 Parallel CUR-Trans with Degree-based Sampling

Lemma. 4.5 *If the sampling ratio satisfies $\rho > \frac{2500\theta^2}{nD}$, then, with probability at least 0.9, the error-bound adjustment induced by degree-aware sampling for CUR-Trans is*

$$\begin{aligned} \Delta \mathcal{E}_{CUR-D}^{(D)} &= (1 - N\alpha_r)^{1/4} \frac{\sqrt{50}\alpha \|A\|_F}{1 - \alpha} \left[(1 - N\beta_c)^{1/4} - (1 - N\alpha_c)^{1/4} \right] \\ &+ (1 - N\alpha_r)^{1/2} \frac{50\alpha \|A\|_F^2}{(1 - \alpha)\sigma_{\min}(W)} \left[(1 - N\beta_c)^{1/2} - (1 - N\alpha_c)^{1/2} \right], \end{aligned}$$

where $\beta_c = 1/\bar{d}$, and $\bar{d}$ is the average degree of graph G .

Proof. All proofs follow the definitions and procedures of Lemma 3.1, with the column sampling distribution replaced by sampling according to the degree, while the row sampling distribution remains unchanged. Each row is sampled independently with probability p_i , and each column is sampled independently with probability q_j , where

$$p_i = \frac{\|A_{i,:}\|_2^2}{\|A\|_F^2}, \quad q_j = \frac{d_j}{\sum_j d_j}.$$

Then, we have

$$\mathbb{E}[\|A_{\mathcal{R},C^c}\|_F^2] \leq \mathbb{E}[\|A_{:,C^c}\|_F^2] = \sum_j \|A_{:,j}\|_2^2 (1 - cq_j) = \|A\|_F^2 - c \sum_j \frac{d_j}{2m} \frac{1}{d_j} = \|A\|_F^2 - \frac{cn}{2m} = (1 - c\beta_c) \|A\|_F^2,$$

where $\beta_c = \frac{1}{\bar{d}}$, $\bar{d}$ is the average degree of graph, m is the total number of edges of graph G , and n is the total number of vertices of graph G . The error bound for $\|A_{\mathcal{R}^c,C}\|_F$ remains unchanged.

$$\mathbb{E}[\|A_{\mathcal{R}^c,C}\|_F^2] \leq (1 - r\alpha_r) \|A\|_F^2.$$

Similarly, the corresponding modifications are made.

$$\mathbb{E}[\|A_{\mathcal{R},C^c}\|_F^2] \leq \sqrt{(1 - c\beta_c)(1 - r\alpha_r)} \|A\|_F^2.$$

By applying Markov's inequality, the expectation in the above inequality is removed. With probability at least $(1 - \delta)^3 > 94\%$, the error bound holds.

$$\begin{aligned} \|A - CUR\|_F &\leq \|A_{\mathcal{R}^c,C^c}\|_F + \frac{\|A_{\mathcal{R}^c,C}\|_F \|A_{\mathcal{R},C^c}\|_F}{\sigma_{\min}(W)} \\ &\leq 5\sqrt{2}(1 - c\beta_c)^{1/4} (1 - r\alpha_r)^{1/4} \|A\|_F + \frac{\left(5\sqrt{2}(1 - c\beta_c)^{1/2} \|A\|_F\right) \left(5\sqrt{2}(1 - r\alpha_r)^{1/2} \|A\|_F\right)}{\sigma_{\min}(W)} \\ &= 5\sqrt{2} \left(1 - \frac{\rho n \alpha_r}{D}\right)^{1/4} \left(1 - \frac{\rho n}{D\bar{d}}\right)^{1/4} \|A\|_F + \frac{50 \left(1 - \frac{\rho n \alpha_r}{D}\right)^{1/2} \left(1 - \frac{\rho n}{D\bar{d}}\right)^{1/2}}{\sigma_{\min}(W)} \|A\|_F^2 \end{aligned} \quad (19)$$

$$(20)$$

Finally, we supplement the discussion on the the row and column spaces of W sampled from R and C . The row sampling remains unchanged, and $r = 2500\theta^2$ still ensures that $\text{Row}(W) = \text{Row}(C)$ holds. When the column sampling is changed and $c = \frac{2500\theta^2}{\varrho_r}$, we have $\text{Column}(R) = R$, where $\varrho_r = \min_j \left\{ \frac{d_j}{\sum_j d_j} \cdot \frac{\|A\|_F^2}{\|A_{:,j}\|_2^2} \right\}$. Moreover, assuming that both conditions are desired to hold with probability 98%, it then follows that the above error bound inequality ?? holds with probability at least $(1 - \delta)^5 > 90\%$. Lastly, the lemma is proved. $\square$

Noted: As long as the graph is not regular, $(1 - c/\bar{d}) \leq (1 - \rho)$, and the difference is more pronounced in sparse graphs. Therefore, replacing the column sampling with degree-based sampling is preferable.

A supplementary note on why the row sampling is not also replaced by a degree-based sampling distribution: if the row sampling were also changed to degree-based, then the above α_r would need to be replaced by $\beta_r = \frac{\sum_i \Delta_i / d_i^2}{2m \cdot \|A\|_F^2}$, where $\Delta_i = \sum_{j \in \mathcal{N}(i)} d_j$ (the sum of the degrees of the neighbors of vertex i .)

$$\mathbb{E}[\|A_{\mathcal{R}^c,C}\|_F^2] \leq \mathbb{E}[\|A_{\mathcal{R}^c,:}\|_F^2] = \|A\|_F^2 - \frac{r}{2m} \sum_i \frac{\Delta_i}{d_i^2}.$$

Moreover, we have

$$\begin{aligned} d_{\min} \cdot d_i \leq \Delta_i \leq d_{\max} \cdot d_i &\implies \frac{d_{\min}}{d_i} \leq \frac{\Delta_i}{d_i^2} \leq \frac{d_{\max}}{d_i}. \\ d_{\min} \sum_i \frac{1}{d_i} \leq \sum_i \frac{\Delta_i}{d_i^2} \leq d_{\max} \sum_i \frac{1}{d_i} &\implies d_{\min} \|A\|_F^2 \leq \sum_i \frac{\Delta_i}{d_i^2} \leq d_{\max} \|A\|_F^2 \\ \frac{d_{\min}}{2m} \leq \beta_r \leq \frac{d_{\max}}{2m} &\implies \frac{d_{\min}}{n\bar{d}} \leq \beta_r \leq \frac{d_{\max}}{n\bar{d}} \end{aligned}$$

Therefore, the upper bound of β_r is larger than α_r .

Finally, we add a supplementary remark. Although a larger number of samples r and c is required for the rigor of the proof to guarantee the error bound, since the row sampling remains unchanged, we only need to discuss the column sampling number c . In practice, only when linearly independent columns are sampled can the column space be preserved. For the transition matrix of an undirected graph, two linearly dependent columns correspond to two vertices that connect to the same set of neighboring vertices. When sampling according to ℓ_2^2 -norm, the column sampling tends to favor vertices with lower degrees, whereas degree-based sampling tends to favor vertices with higher degrees. In real-world graphs, vertices with lower degrees are actually more likely to have linearly dependent columns. Therefore, degree-based sampling can often achieve better subspace preservation with fewer sampled rows and columns.

8 Lemma 4.6 Parallel T^2 -Approx with Degree-based Sampling

Lemma. 4.6 *If $\rho > 0$, then, with probability at least 0.9, the error-bound adjustment induced by degree-aware sampling for T^2 -Approx is*

$$\Delta \mathcal{E}_{T^2-D}^{(D)} = \left[(1 - N\beta_{cr})^{1/2} - (1 - N\alpha_{cr})^{1/2} \right] \frac{\sqrt{10}\alpha \|A\|_F^2}{2n(1 - \alpha)},$$

where $\beta_{cr} = (\sum_i \|A_{i,:}\|_2) / (\|A\|_F^2 \sum_k d_k \|A_{k,:}\|_2)$.

Proof. All proofs follow the definitions and procedures of Lemma 3.1 and 4.3, with the column sampling distribution replaced by sampling according to the degree, while the row sampling distribution remains unchanged. The sampling probabilities of vertex i being included as a column in C and as a row in R are given by:

$$p_i = \frac{d_i \|A_{i,:}\|_2}{\sum_{k=1}^n d_k \|A_{k,:}\|_2}.$$

Define the parameter β_{rd} .

$$\beta_{rd} = \frac{\sum_i d_i \|A_{i,:}\|_2 \|A_{:,i}\|_2^2}{(\sum_k d_k \|A_{k,:}\|_2) \|A\|_F^2}.$$

Because $\|A_{:,i}\|_2^2 = \frac{1}{d_i}$, we have

$$\beta_{rd} = \frac{\sum_i d_i \|A_{i,:}\|_2 \frac{1}{d_i}}{(\sum_k d_k \|A_{k,:}\|_2) \|A\|_F^2} = \frac{\sum_i \|A_{i,:}\|_2}{(\sum_k d_k \|A_{k,:}\|_2) \|A\|_F^2} = \frac{\sum_i \|A_{i,:}\|_2}{(\sum_k d_k \|A_{k,:}\|_2) \sum_k d_k^{-1}} \geq \frac{1}{d_{\max} \sum_k d_k^{-1}} \geq \frac{d_{\min}}{nd_{\max}}.$$

The remaining procedures are the same as the proof of Lemma 3.2, except that α_{rd} is replaced by β_{rd} . Therefore, based on Markov's inequality, the following inequality holds with probability at least 90%.

$$\begin{aligned} \|A^2 - XY\|_F &\leq \sqrt{10}(1 - c\beta_{rd})^{1/2} \|A\|_F^2 \\ \|A^2 - XY\|_F &\leq \sqrt{10} \left(1 - \frac{cd_{\min}}{nd_{\max}} \right)^{1/2} \|A\|_F^2. \end{aligned} \tag{21}$$

Lastly, the lemma is proved. $\square$

$$\|A^2 - XY\|_F \leq \sqrt{10} \left(1 - \frac{\rho n \beta_{rd}}{D} \right)^{1/2} \|A\|_F^2$$

$\square$

Noted: If the entire distribution is replaced by sampling completely according to the degree, we only need to replace β_{rd} with $\gamma_{cr} = \frac{n}{2m\|A\|_F^2} = \frac{1}{d\|A\|_F^2}$.

9 Lemma 4.7 Parallel T^2 -Approx with Vertex Aggregation

Lemma. 4.8 For $\rho > 0$, vertex-level aggregation does not adjust the error bound for T^2 -Approx:

$$\Delta \mathcal{E}_{T^2-A}^{(D)} = 0.$$

Proof. For the T^2 -Approx approximation, the single-vertex estimation error of vertex i is jointly determined by all unsampled vertices and is not directly related to whether the corresponding row or column of vertex i is sampled. Let Ω denote the set of sampled vertices. Then,

$$\begin{aligned} A^2 - CR &= A^2 - ASA = A(I - S)A \\ (A^2 - CR)_{ij} &= \sum_{k=1}^n A_{ik}(I - S)_{kk}A_{kj} = \sum_{k \notin \Omega} A_{ik}A_{kj}. \end{aligned}$$

Therefore, the error of the i -th row is

$$\begin{aligned} \|(A^2)_{i,:} - (CR)_{i,:}\|_2 &= \|A_{i,:}(I - S)A\|_2 \\ &= \left\| \sum_{k \notin \Omega} A_{ik}A_{k,:} \right\|_2. \end{aligned}$$

Similarly, the error of the i -th column is

$$\begin{aligned} \|(A^2)_{:,i} - (CR)_{:,i}\|_2 &= \|A(I - S)A_{:,i}\|_2 \\ &= \left\| \sum_{k \notin \Omega} A_{:,k}A_{ki} \right\|_2. \end{aligned}$$

The above results show that the errors of both the i -th row and the i -th column are determined by the accumulated contributions from all unsampled vertices. Therefore, whether vertex i itself is sampled into R or C is not a direct factor determining its T^2 approximation error. This indicates that the single-vertex estimation error of CR as an approximation of T^2 is essentially determined by the accumulated errors introduced by unsampled vertices acting as intermediate states in the second-order propagation, rather than by whether vertex i itself is included in R or C .

Due to shuffled partitioning, the set of unsampled vertices in each sample is expected to be equivalent. Therefore, for T^2 -Approx, the Vertex-Level aggregation degenerates into uniform aggregation in expectation. $\square$

10 Lemma 4.8 Parallel CUR-Trans with Vertex-Level Aggregation

Lemma. 4.7 If the sampling ratio satisfies $\rho > \frac{2500\theta^2}{nD}$, then, with probability at least 0.9, the error-bound adjustment induced by vertex-level aggregation for CUR-Trans is

$$\Delta \mathcal{E}_{CUR-A}^{(D)} = -0.5\alpha^3(1 - \alpha)n^2N^2\lambda^2(1 + \eta)^2 \left(\mathcal{E}_{CUR}^{(D)} \right)^2,$$

$$\text{where } \eta = \frac{\sum_i d_i \|A_{i,:}\|_2^2}{2m\|A\|_F^2}.$$

Proof. First, we identify the component that determines the single-vertex PageRank error. Let $\tilde{A} = CUR$. The PageRank vector satisfies:

$$(\mathbf{I} - \alpha A)\pi = (1 - \alpha) \left[\frac{1}{n} \right]_{n \times 1}, \quad (\mathbf{I} - \alpha \tilde{A})\tilde{\pi} = (1 - \alpha) \left[\frac{1}{n} \right]_{n \times 1}.$$

Then, we have

$$\begin{aligned}
(\mathbf{I} - \alpha \tilde{A})\tilde{\pi} - (\mathbf{I} - \alpha A)\pi &= \mathbf{0} \\
(\mathbf{I} - \alpha A)(\tilde{\pi} - \pi) &= \alpha (A - \tilde{A}) \tilde{\pi} \\
\tilde{\pi} - \pi &= \alpha (\mathbf{I} - \alpha A)^{-1} (A - \tilde{A}) \tilde{\pi} \\
\|\tilde{\pi} - \pi\|_1 &\leq \alpha \|\tilde{\pi}\|_1 \|A - \tilde{A}\|_1 \|(\mathbf{I} - \alpha A)^{-1}\|_1 \\
\|\tilde{\pi} - \pi\|_1 &\leq \alpha \left(\sum_{k=0}^{\infty} \alpha^k A^k \right) \|A - \tilde{A}\|_1 \\
\|\tilde{\pi} - \pi\|_1 &\leq \frac{\alpha}{1 - \alpha} \|A - \tilde{A}\|_1
\end{aligned}$$

Define $\Delta = \tilde{A} - A$, and compute the error of each vertex.

$$\begin{aligned}
\tilde{\pi} - \pi &= \alpha \Delta \tilde{\pi} + \alpha A (\tilde{\pi} - \pi) \\
(\tilde{\pi} - \pi)_i &= \alpha (\Delta \tilde{\pi})_i + \alpha (A (\tilde{\pi} - \pi))_i.
\end{aligned}$$

Then, we have

$$\begin{aligned}
|\tilde{\pi}(i) - \pi(i)| &= |\alpha (\Delta \tilde{\pi})_i + \alpha (A (\tilde{\pi} - \pi))_i| \\
&= \left| \alpha \sum_{j=1}^n \tilde{\pi}_j \Delta_{i,j} + \alpha \sum_{j=1}^n A_{i,j} (\tilde{\pi}_j - \pi_j) \right| \\
&\leq \alpha \sum_{j=1}^n \tilde{\pi}_j |\Delta_{i,j}| + \alpha \sum_{j=1}^n A_{i,j} |\tilde{\pi}_j - \pi_j| \\
&\leq \alpha \|\tilde{\pi}\|_1 \|\Delta_{i,:}\|_1 + \alpha \|A_{i,:}\|_1 \|\tilde{\pi} - \pi\|_1 \\
&\leq \alpha \|\Delta_{i,:}\|_1 + \alpha \|A_{i,:}\|_1 \|\tilde{\pi} - \pi\|_1 \\
&\leq \alpha \|\Delta_{i,:}\|_1 + \frac{\alpha^2 \cdot \|A_{i,:}\|_1}{1 - \alpha} \|\Delta\|_1.
\end{aligned}$$

Therefore, it can be observed that the single-vertex PageRank error can be controlled by a linear combination of the global error $\|A - \tilde{A}\|$ and the corresponding row-wise error of $\|A - \tilde{A}\|$. Let the global term $\mathcal{G} = \frac{\alpha^2}{1-\alpha} \sqrt{n} \|\Delta\|_F \|A_{i,:}\|_1$ be identical for all vertices, serving as the "base" error component, while the local term varies with the vertex state and determines the differences in errors among different vertices.

Analyze $\|\Delta_{i,:}\|_1$ under different cases. Let $\mathcal{R}$ denote the set of sampled rows and $\mathcal{C}$ denote the set of sampled columns. Define $W = A_{\mathcal{R},\mathcal{C}}$, $P_{\text{col}} = WW^+$ (the orthogonal projection onto the column space of W), and $P_{\text{row}} = W^+W$ (the orthogonal projection onto the row space of W).

$$A = \begin{bmatrix} A_{\mathcal{R},\mathcal{C}} & A_{\mathcal{R},\mathcal{C}^c} \\ A_{\mathcal{R}^c,\mathcal{C}} & A_{\mathcal{R}^c,\mathcal{C}^c} \end{bmatrix}, \quad R = S_r A = \begin{bmatrix} A_{\mathcal{R},\mathcal{C}} & A_{\mathcal{R},\mathcal{C}^c} \\ 0 & 0 \end{bmatrix}, \quad C = AS_c = \begin{bmatrix} A_{\mathcal{R},\mathcal{C}} & 0 \\ A_{\mathcal{R}^c,\mathcal{C}} & 0 \end{bmatrix}.$$

Obtain the exact block structure of Δ .

$$\Delta = \begin{bmatrix} 0 & (I - P_{\text{col}})A_{\mathcal{R},\mathcal{C}^c} \\ A_{\mathcal{R}^c,\mathcal{C}}(I - P_{\text{row}}) & A_{\mathcal{R}^c,\mathcal{C}^c} - A_{\mathcal{R}^c,\mathcal{C}}W^+A_{\mathcal{R},\mathcal{C}^c} \end{bmatrix}$$

For a vertex i , there are four scenarios in total, corresponding to the single-row error in each case. **Scenario 1:** $i \in \mathcal{R}$ and $i \in \mathcal{C}$. We have that

$$\begin{aligned}
\tilde{A}_{i,:} &= [A_{i,\mathcal{C}}W^+W \quad A_{i,\mathcal{C}}W^+A_{\mathcal{R},\mathcal{C}^c}] = [A_{i,\mathcal{C}} \quad A_{i,\mathcal{C}}W^+A_{\mathcal{R},\mathcal{C}^c}] \\
\implies \|A_{i,:} - \tilde{A}_{i,:}\|_2^2 &= \|A_{i,\mathcal{C}^c} - A_{i,\mathcal{C}}W^+A_{\mathcal{R},\mathcal{C}^c}\|_2^2 = \|(I - P_{\text{col}})A_{\mathcal{R},\mathcal{C}^c}\|_2^2.
\end{aligned}$$

Scenario 2: $i \in \mathcal{R}$ but $i \notin \mathcal{C}$. Since the i -th row is sampled but the i -th column is not sampled, we have $A_{i,\mathcal{C}}W^+W = A_{i,\mathcal{C}}$, which is the same as Scenario 1.

$$\|\tilde{A}_{:,j} - A_{:,j}\|_2^2 = \|(I - P_{\text{col}})A_{\mathcal{R}^c,\mathcal{C}}\|_2^2.$$

Scenario 3: $i \notin \mathcal{R}$ but $i \in \mathcal{C}$. Since the i -th row is not sampled but the i -th column is sampled, $A_{i,C}$ is not a row of W . Therefore,

$$\begin{aligned}\tilde{A}_{i,C} &= A_{i,C}W^+W = A_{i,C}P_{\text{row}} \\ \implies \|\tilde{A}_{i,:} - A_{i,:}\|_2^2 &= \|A_{i,C}(I - P_{\text{row}})\|_2^2 + \|A_{i,C^c} - A_{i,C}W^+A_{\mathcal{R},C^c}\|_2^2.\end{aligned}$$

Scenario 4: $i \notin \mathcal{R}$ but $i \notin \mathcal{C}$. This Scenario is the same as Scenario 3.

$$\|\tilde{A}_{i,:} - A_{i,:}\|_2^2 = \|A_{i,C}(I - P_{\text{row}})\|_2^2 + \|A_{i,C^c} - A_{i,C}W^+A_{\mathcal{R},C^c}\|_2^2.$$

In summary, $\Delta_{i,:}$ only depends on whether the corresponding row of vertex i is sampled into R and is independent of the column sampling condition. If $i \in \mathcal{R}$ and $\rho \geq \frac{2500\theta^2}{Dn\varrho^2}$, with probability $0.98 * 0.98$ we have with

$$\mathbb{E}[\|\tilde{A}_{i,:} - A_{i,:}\|_2] = \|(I - P_{\text{col}})A_{\mathcal{R},C^c}\|_2^2 = 0.$$

If $i \notin \mathcal{R}$, we have

$$\begin{aligned}\|\tilde{A}_{i,:} - A_{i,:}\|_2 &= \|A_{i,C}(I - P_{\text{row}})\|_2 + \|A_{i,C}W^+A_{\mathcal{R},C^c} - A_{i,C^c}\|_2 \\ &= \|A_{i,C}W^+A_{\mathcal{R},C^c} - A_{i,C^c}\|_2 \\ &\leq \|A_{i,C^c}\|_2 + \frac{\|A_{i,C}\|_2 \cdot \|A_{\mathcal{R},C^c}\|_2}{\sigma_{\min}(W)}.\end{aligned}$$

At last, we analyze the PageRank error of aggregating D samples for a single vertex i . Since $\alpha\|\Delta_{i,:}\|_1$ differs depending on whether vertex i is sampled as a row into R , we use Φ_R^c to denote the error contribution of this term when vertex i is not sampled as a row into R , and Φ_R to denote the error contribution of this term when vertex i is sampled as a row into R . And, $\Phi_R = 0$.

$$|\pi(v) - \tilde{\pi}(v)| = \sum_d \omega_d(v) |\pi_d(v) - \tilde{\pi}_d(v)|.$$

Case 1: If no sample contains the row or column associated with i , the aggregation degenerates into uniform aggregation. Therefore, the single-vertex error is:

$$|\pi(i) - \tilde{\pi}(i)| \leq \mathcal{G} + \Phi_R^c.$$

Case 2: If in the sample S_{d^*} , the column-sampled submatrix C_{d^*} contains the column associated with i , but the row-sampled submatrix R_{d^*} does not contain the row associated with i , then the single-vertex error is:

$$|\pi(i) - \tilde{\pi}(i)| = \sum_{d \neq d^*} \frac{1 - \lambda/2}{D - 1} |\pi_d(v) - \tilde{\pi}_d(v)| + \sum_{d=d^*} \frac{\lambda}{2} |\pi_d(v) - \tilde{\pi}_d(v)| \leq \mathcal{G} + \Phi_R^c.$$

If in the sample S_{d^*} , the row-sampled submatrix C_{d^*} contains the column associated with i , but the column-sampled submatrix R_{d^*} does not contain the row associated with i , then the single-vertex error is:

$$|\pi(i) - \tilde{\pi}(i)| = \sum_{d \neq d^*} \frac{1 - \lambda/2}{D - 1} |\pi_d(v) - \tilde{\pi}_d(v)| + \sum_{d=d^*} \frac{\lambda}{2} |\pi_d(v) - \tilde{\pi}_d(v)| \leq \mathcal{G} + (1 - \lambda/2) \cdot \Phi_R^c.$$

Case 3: If in the sample S_{d^*} , both the column-sampled submatrix C_{d^*} and row-sampled submatrix R_{d^*} contain the column and row associated with i , then the single-vertex error is:

$$|\pi(i) - \tilde{\pi}(i)| = \sum_{d \neq d^*} \frac{1 - \lambda}{D - 1} |\pi_d(v) - \tilde{\pi}_d(v)| + \sum_{d=d^*} \frac{\lambda}{2} |\pi_d(v) - \tilde{\pi}_d(v)| \leq \mathcal{G} + (1 - \lambda) \cdot \Phi_R^c.$$

Calculate the expectation

$$\begin{aligned}\mathbb{E}[|\pi(i) - \tilde{\pi}(i)|] &\leq \mathcal{G} + \sum_i (1 - rp_i) \cdot \Phi_R^c + [rp_i(1 - cq_i)(1 - \lambda/2) + rp_i cq_i(1 - \lambda)] \cdot \Phi_R^c \\ &= \mathcal{G} + \sum_i (1 - rp_i) \cdot \Phi_R^c + rp_i(1 - \lambda/2 - cq_i \cdot \lambda/2) \cdot \Phi_R^c \\ &= \mathcal{G} + \Phi_R^c - \sum_i rp_i(\lambda/2 + cq_i \cdot \lambda/2) \cdot \Phi_R^c \\ &\leq \mathcal{G} + \left(1 - \frac{r\lambda}{2} - rc\lambda\eta\right) \Phi_R^c.\end{aligned}$$

where $\eta = \frac{\sum_i d_i \|A_{i,:}\|_2^2}{2m\|A\|_F^2}$. For uniform weighted aggregation,

$$\mathbb{E} [|\pi(i) - \tilde{\pi}(i)|] \leq \mathcal{G} + \Phi_R^c.$$

Therefore, $\Delta \mathcal{E}_{\text{CUR-A}}^{(D)}$ for each vertex is

$$\begin{aligned} \mathbb{E} \left[\sum_i \Delta \mathcal{E}_{\text{CUR-A}}^{(D)}(i) \right] &= \mathbb{E} \left[\sum_i - \left(\frac{N\lambda}{2} + N^2\lambda\eta \right) \Phi_R^c \right] \\ &= \mathbb{E} \left[\sum_i - \left(\frac{N\lambda}{2} + N^2\lambda\eta \right)^2 \alpha^2 \|\Delta_{i,:}\|_1^2 \right] \\ &= \sum_i - \left(\frac{N\lambda}{2} + N^2\lambda\eta \right)^2 \alpha^2 \mathbb{E} [\alpha^2 n \cdot \|\tilde{A}_{i,:} - A_{i,:}\|_2^2] \\ &= -\alpha^4 n \left(\frac{N\lambda}{2} + N^2\lambda\eta \right)^2 \mathbb{E} [\|\tilde{A} - A\|_F^2] \\ &= -0.5\alpha^3 (1 - \alpha)n^2 N^2 \lambda^2 (1 + \eta)^2 \left(\mathcal{E}_{\text{CUR}}^{(D)} \right)^2 \end{aligned}$$

where $\mathbb{E} [\|\tilde{A} - A\|_F] \leq (1 - \rho n \alpha_c)^{1/4} (1 - \rho n \alpha_r)^{1/4} \|A\|_F + \frac{(1 - \rho n \alpha_r)^{1/2} (1 - \rho n \alpha_c)^{1/2}}{\sigma_{\min}(W)} \|A\|_F^2$

$$\mathbb{E} [\Phi_R^c] = \mathbb{E} [\alpha \|\Delta_{i,:}\|_1] \leq \alpha \sqrt{n} \cdot \mathbb{E} [\|\tilde{A}_{i,:} - A_{i,:}\|_2]$$

$$\begin{aligned} \mathbb{E} [\|\tilde{A}_{i,:} - A_{i,:}\|_2] &\leq \mathbb{E} [\|A_{i,C^c}\|_2] + \frac{\mathbb{E} [\|A_{i,C}\|_2] \cdot \mathbb{E} [\|A_{\mathcal{R},C^c}\|_2]}{\sigma_{\min}(W)} \\ &\leq (1 - c\beta_c)^{1/4} (1 - r\alpha_r)^{1/4} \|A_{i,:}\|_2 + \frac{(1 - c\beta_c)^{1/2} (1 - r\alpha_r)^{1/2} \|A_{i,:}\|_2^2}{\sigma_{\min}(W)} \end{aligned}$$

□

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